

4.1.1. Design and Analysis of an FIR Low-Pass Filter Using the Rectangular Window Technique in MATLAB

```
clc;

clear;

close all;

%% --- Input Parameters ---

fp = 1000; % Passband cutoff (Hz)

fs = 4000; % Sampling frequency (Hz)

N = 50; % Filter order (even number preferred)

wc = fp/(fs/2); % Normalized cutoff frequency (0 to 1)

%% --- 1. FIR Low Pass using Rectangular Window ---

b_rect = fir1(N, wc, 'low', rectwin(N+1));

disp('--- Rectangular Window Coefficients ---');

disp(b_rect');

%% --- 2. FIR Low Pass using Hamming Window ---

b_hamm = fir1(N, wc, 'low', hamming(N+1));

disp('--- Hamming Window Coefficients ---');

disp(b_hamm');

%% --- 3. FIR Low Pass using Hanning Window ---

b_hann = fir1(N, wc, 'low', hanning(N+1));

disp('--- Hanning Window Coefficients ---');

disp(b_hann');

%% --- Plot Frequency Responses ---

figure;

freqz(b_rect,1);

title('Low-pass FIR using Rectangular Window');

figure;

freqz(b_hamm,1);

title('Low-pass FIR using Hamming Window');
```

figure;

```
freqz(b_hann,1);
```

```
title('Low-pass FIR using Hanning Window');
```

OUTPUT;



